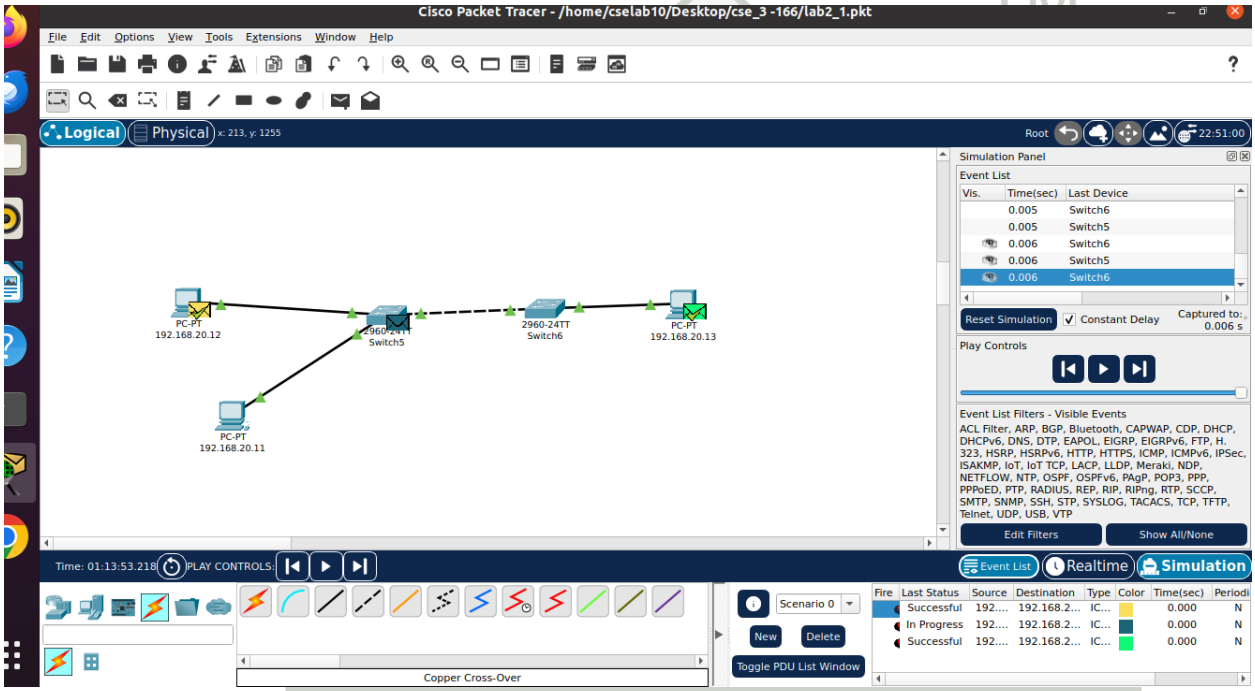
1.Peer to peer (P2P) Topology

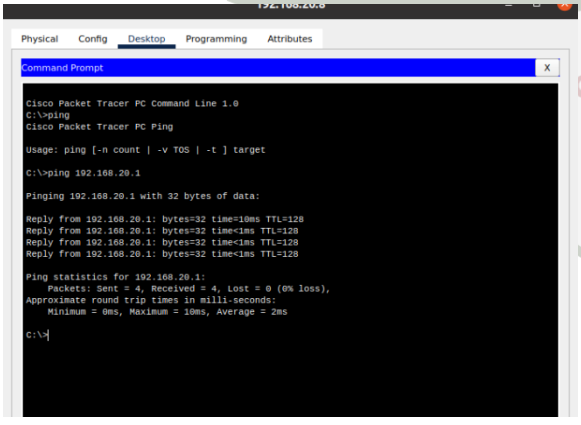
Aim: To establish direct communication between two devices using a dedicated communication link.

Description: Peer-to-Peer topology connects exactly two devices through a single communication channel. It provides fast, secure, and reliable data transmission because the link is dedicated to only those two devices.

Connection:



OUTPUT:




```
C:\>ping 192.168.20.12

Pinging 192.168.20.12 with 32 bytes of data:

Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

JUSTIFICATION:

Peer-to-Peer topology is simple to implement and provides highspeed communication with minimum data loss. It is commonly used to connect two routers or two computers directly

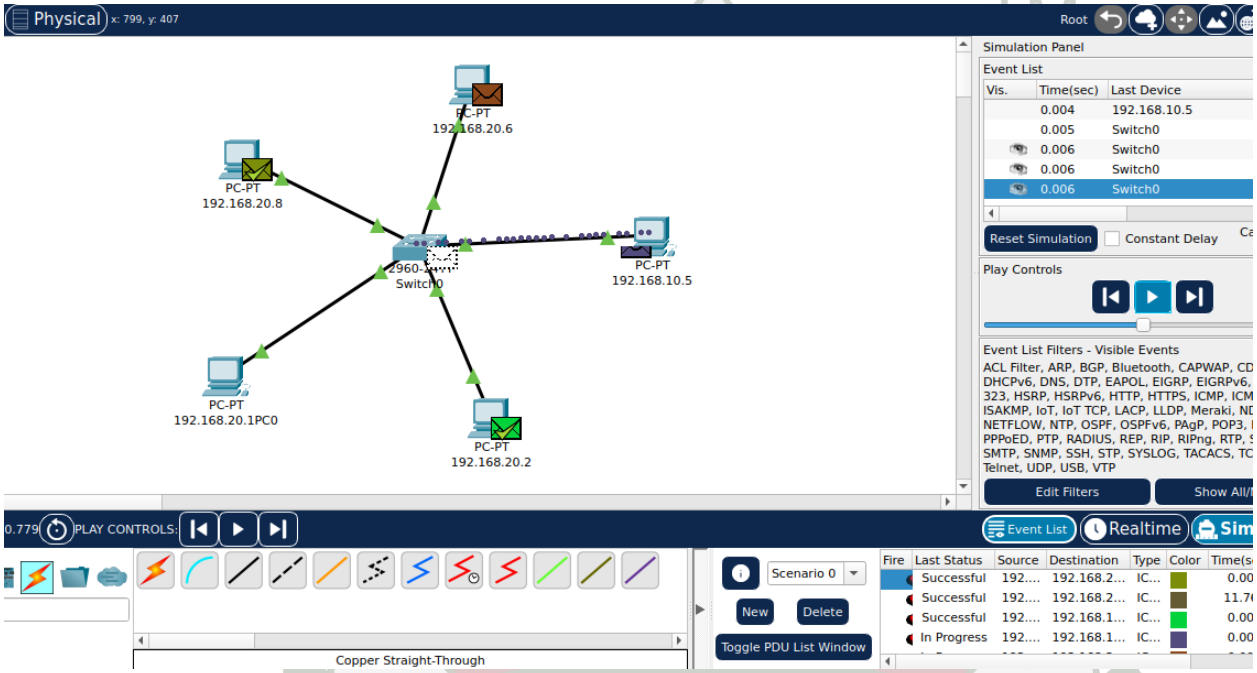
2. Star Topology

Aim

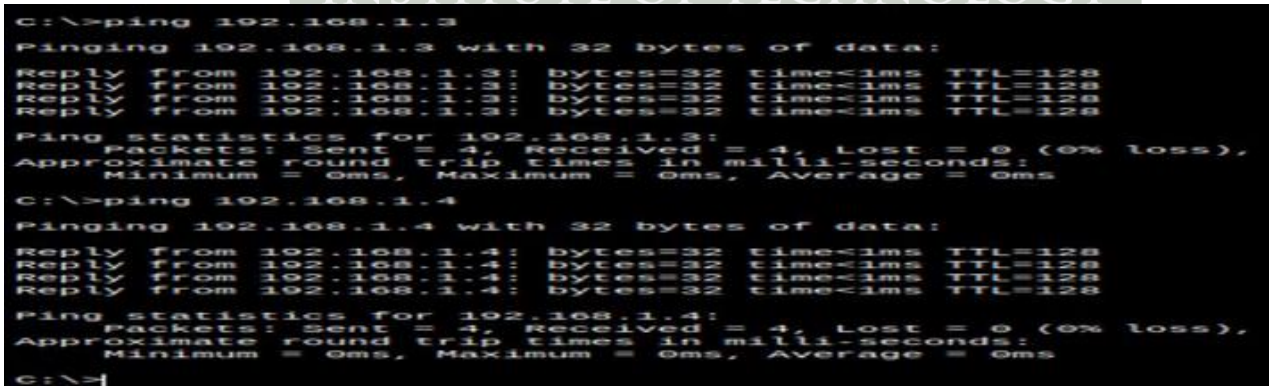
To connect all devices through a central switch or hub.

Description

In Star topology, every device is connected directly to a central switch or hub. All communication between devices passes through the central device.



Output:



Justification

Star topology is easy to install, manage, and troubleshoot. Failure of one cable affects only the corresponding device, making it the most widely used topology in LANs.

3. Mesh Topology

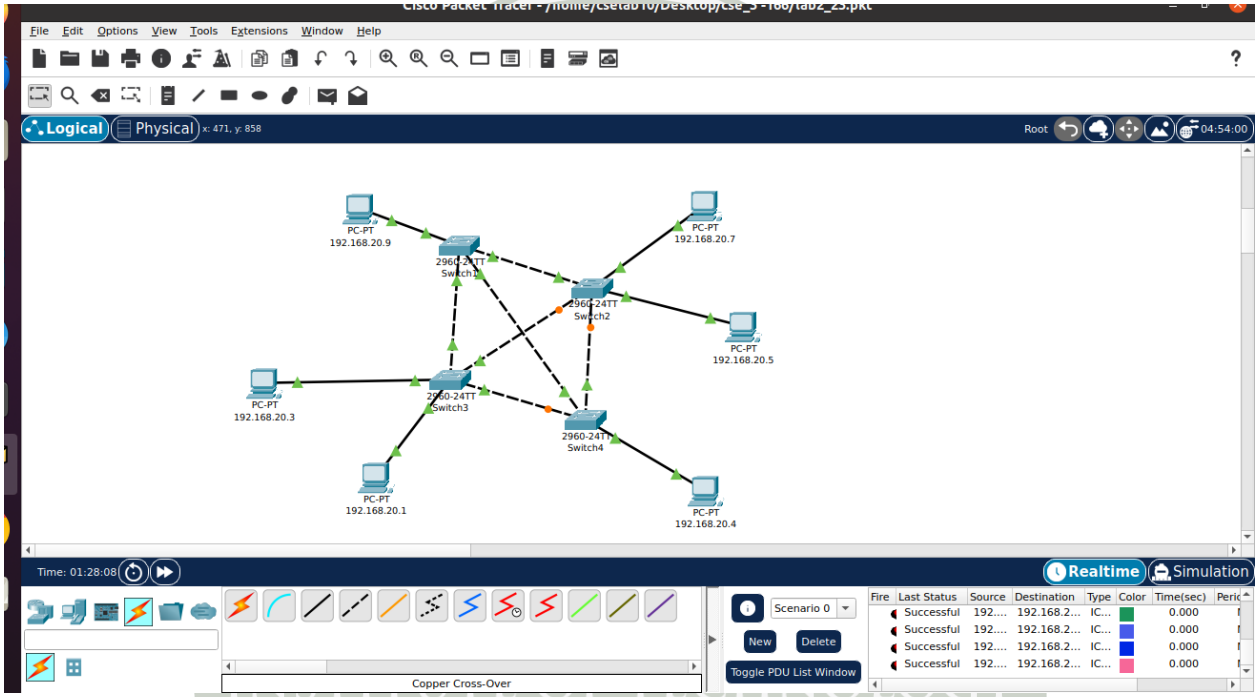
Aim :

To provide multiple communication paths between all devices for maximum reliability.

Description

very device is connected to every other device through dedicated links. This provides multiple paths for data transmission and ensures high reliability.

Connection



Output:

PDU List List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	192.168.20.3	192.168.20.4	ICMP		0.000	N	0	(edit)	
	Successful	192.168.20.7	192.168.20.1	ICMP		0.000	N	1	(edit)	

Justification

Mesh topology offers excellent fault tolerance and network reliability. It is mainly used in critical applications where uninterrupted communication is required.

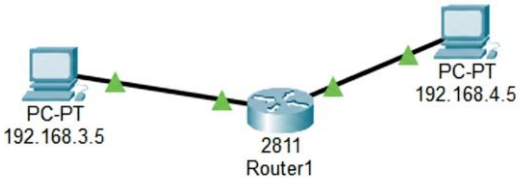
Router Configuration

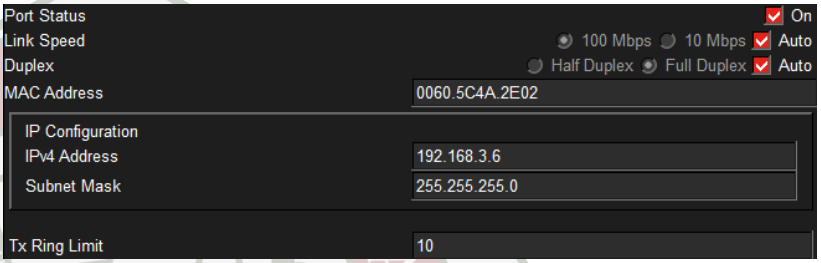
Aim To configure a router and assign IP addresses to its interfaces so that two computers connected through the router can communicate successfully

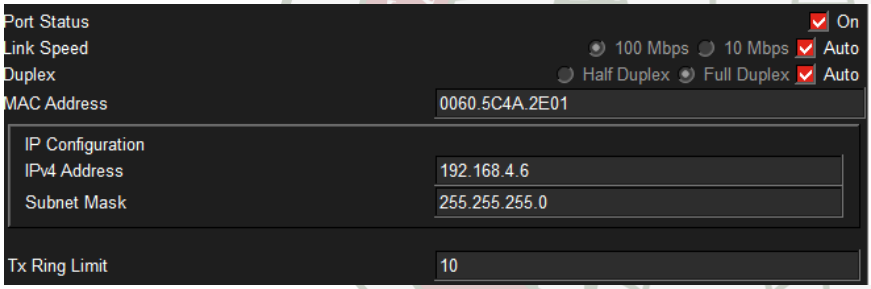
Description

A router is a network device that connects different devices and helps them communicate by sending data to the correct destination.

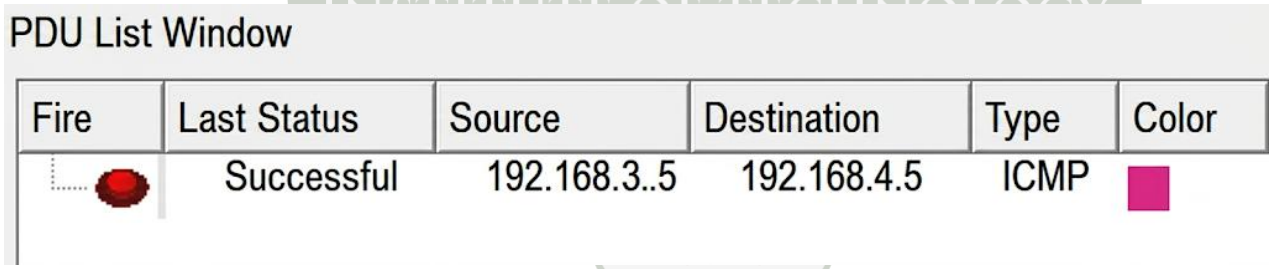
Connection







Output:



Justification

This experiment helps us learn how to configure a router and check whether the connected devices can communicate successfully.